Data portability, file formats and compression

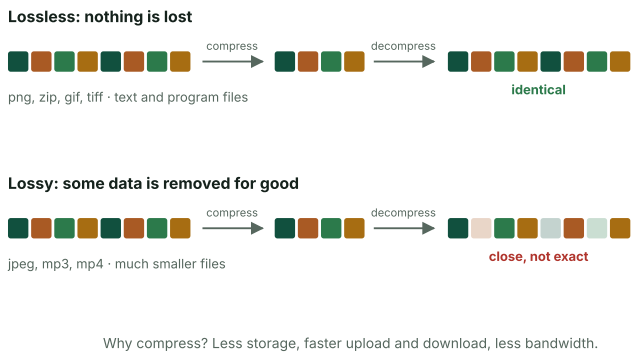
L05
FACT FILE

Specification: students should be able to

- Demonstrate understanding of data portability and the following file formats that support it: jpeg, tiff, png, pict, gif, txt, csv, rtf, mp3, mp4, midi, mpeg, avi, pdf, wav and wma
- Choose and justify a suitable file format for a given purpose
- Demonstrate understanding of the need for data compression
- Describe lossy and lossless compression and give examples of each

CCEA fact file: Digital Data 04: Portability

Key words: data portability, interoperability, file format, file extension, export, import, compression, lossy compression, lossless compression, zip, jpeg, png, gif, tiff, csv, rtf, pdf, midi



Lossless compression restores every block of data; lossy compression permanently removes some

Data portability and file formats

Data portability means being able to move data from one computer, device or application to another without having to type it in again. Mark schemes also accept the words interoperability or compatibility. Portability depends on file formats. A file format is the agreed way that data is laid out inside a file, and the file extension, such as .jpg or .csv, shows which format is used. If data is saved in a standard format, many different programs on many different systems can open it. For example, a youth football club in Lisburn might keep its members in a database, export them as a csv file and send that file to the league, which opens it in a spreadsheet. Nothing is retyped, so time is saved and no new errors are introduced. Choosing the right format matters. If you pick a format that the other person cannot open, or one that loses quality or formatting, the data is not truly portable.

Image, text and document formats

Jpeg is the usual format for photographs. It uses lossy compression, and the user can choose how much, so files are small and load quickly on web pages. Png uses lossless compression, keeps every detail and supports transparent backgrounds, which makes it good for logos. Gif can only use 256 colours, but it is lossless and can store a series of images that play one after another as a simple animation. Tiff keeps very high quality and moves easily between Windows and Apple computers, so it is used for printing and scanning, but files are large. Pict is Apple's own graphics format and can hold bitmap and vector images. For text, txt holds plain text with no formatting at all. Csv stores table data with commas between the fields. Rtf keeps basic formatting such as fonts and bold and can be opened by most word processors. Pdf, created by Adobe, keeps fonts, images and layout exactly the same on every device, and is usually read-only.

Sound and video formats

Wav stores sound without any compression, so the quality is high but the files are large, which makes it suitable for recording and editing. Mp3 compresses sound using lossy compression. It removes sounds that most people would not hear, giving a much smaller file that still sounds close to the original, so it is used for music and podcasts. Wma is Microsoft's compressed audio format, made for Windows Media Player. Midi is different from all the others. It does not store recorded sound at all. Instead it stores instructions such as which notes to play, on which instrument and for how long, so the files are tiny and can be passed between electronic keyboards and computers. For video, mp4 compresses both the pictures and the sound and can also hold subtitles and still images; it is widely used for streaming. Mpeg is named after the group of experts who created standards for compressing video. Avi is a Microsoft format that keeps audio and video in sync, but files are often large.

Data compression

Compression reduces the size of a file. Smaller files take up less storage space and can be uploaded, downloaded and emailed more quickly, which is important for web pages, streaming and attachments with size limits. There are two types. Lossy compression permanently removes some of the data, such as fine detail or colours that people are unlikely to notice. Files become much smaller, but some quality is lost and the original can never be fully recovered. Jpeg, mp3 and mp4 use lossy compression, and it suits photos, music and video. Lossless compression makes the file smaller without throwing any data away, so when the file is decompressed it is exactly the same as the original. Zipping a folder with a program such as WinZip is lossless, and so is png. Lossless compression does not shrink files as much, but it must be used for text documents, spreadsheets and programs, because losing even a small part would change the data or stop it working.

T Key terms from the mark schemes

Data portability
Being able to move data between computers, devices or software applications without typing it in again, usually by saving it in a standard file format.
Credited: interoperability; compatibility (U1 2021 Q9, 2022 Q11, 2025 Q3)

Compression
Reducing the size of a file so it needs less storage and can be uploaded or downloaded more quickly.
Credited: U1 2022 Q1, 2024 Q7

Lossless compression
Compression that reduces file size without losing any data, so the file can be restored exactly as it was, e.g. zip, png.
Credited: U1 2019 Q4, 2021 Q10, 2023 Q2

PDF
Portable Document Format, developed by Adobe: keeps a document's fonts, images and layout the same on any device; usually read-only.
Credited: U1 2025 Q3

TIFF
Tagged Image File Format: stores high-quality bitmap images that can move between Windows and Apple computers; files are larger than jpegs.
Credited: U1 2022 Q11

MIDI
Musical Instrument Digital Interface: stores and exchanges musical instructions (notes, instruments, timing) between electronic instruments and computers, not recorded sound.
Credited: U1 2021 Q10

File format
The standard way data is organised inside a file, shown by its file extension (e.g. .jpg, .csv), which determines which programs can open it.

Lossy compression
Compression that reduces file size by permanently removing some data, so some quality is lost and the original cannot be fully restored, e.g. jpeg, mp3.
Credited: U1 2019 Q4, 2021 Q10, 2023 Q2

CSV
Comma-separated values: a text format in which each line is a record and the fields are separated by commas; used to move table data between spreadsheets and databases.
Credited: U1 2021 Q2, 2022 Q10

JPEG
Image format using lossy compression whose level the user can control; small files, widely used for photographs on web pages.
Credited: U1 2021 Q10, 2022 Q11, 2025 Q3

WAV
A standard uncompressed audio format; high quality but large files.
Credited: U1 2025 Q3

RTF
Rich Text Format: a text document format with basic formatting that can be opened by different word processors and operating systems.

Exam focus

File formats appear almost every year. Multiple-choice questions on image and video formats came up in 2019 Q1(a)–(c), and 2021 Q10(a) and 2025 Q3(b) asked for formats and their uses. 2022 Q11 asked about portability and compared jpeg with tiff: credit came from saying jpeg uses lossy compression and gives small files for the web, while tiff keeps high quality for printing but makes large files. 2021 Q2(e) tested exporting database data to csv. The Chief Examiner reported that portability (2025 Q3(a)) was poorly answered, so learn a precise definition: moving data between computers or applications without re-entering it. For compression (2021 Q10(a)(ii), 2023 Q2(f), 2024 Q7(a)), give a reason such as faster upload or less storage, and when comparing lossy and lossless always state that lossy loses data permanently while lossless can be restored exactly. When asked to choose a format, name it and justify it with a feature that fits the task.

Unit 1 2025 Q3	[5]	Data portability and file formats
Unit 1 2024 Q7	[5]	Compression, video formats, bit depth
Unit 1 2023 Q2	[10]	Bitmap vs vector, resolution, buffering, sample rate, ADC, compression
Unit 1 2022 Q11	[4]	Data portability; JPEG and TIFF
Unit 1 2022 Q1	[9]	MCQ: storage, bit rate, real-time, compression, buffering, ADC, big data, backup

? Check yourself

1. What is meant by data portability? [2]
2. Name a suitable file format for each: (a) a photo for a website (b) a document that must look the same on every device (c) data exported from a database to a spreadsheet (d) an animated image with few colours. [4]
3. Describe the difference between a midi file and an mp3 file. [2]

Answers: 1. Being able to move data between computers / devices / software applications (1) / without it needing to be typed in again (1). 2. (a) jpeg (accept png) (1) / (b) pdf (1) / (c) csv (1) / (d) gif (1). 3. A midi file stores instructions / musical data such as notes, instrument and timing (1) / an mp3 stores recorded sound in compressed form (1). Accept midi files are much smaller.